

Classification Metrics — Step-by-Step

How to use this PDF: First understand Actual vs Predicted. Then identify TP, TN, FP, FN. For every metric, the **formula is shown first**, followed by substitution, calculation, and final answer.

1. Actual vs Predicted — The Foundation

Actual = what is really true in the real world.

Predicted = what the ML model says.

Positive = the class selected as positive. It can mean disease, spam, fraud, default, pass, etc.

Negative = the other class.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

TP: Actually positive and predicted positive.

FN: Actually positive but predicted negative.

FP: Actually negative but predicted positive.

TN: Actually negative and predicted negative.

Memory rule: Read **Actual first** → **Prediction second**. TP and TN are correct predictions; FP and FN are incorrect predictions.

2. All Formulas

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Error Rate} = (FP + FN) / (TP + TN + FP + FN) = 1 - \text{Accuracy}$$

$$\text{Precision (PPV)} = TP / (TP + FP)$$

$$\text{Recall / Sensitivity / TPR} = TP / (TP + FN)$$

$$\text{Specificity / TNR} = TN / (TN + FP)$$

$$\text{False Positive Rate (FPR), } \alpha = FP / (FP + TN) = 1 - \text{Specificity}$$

$$\text{False Negative Rate (FNR), } \beta = FN / (FN + TP) = 1 - \text{Sensitivity}$$

$$\text{Power} = \text{Sensitivity} = \text{TPR} = 1 - \beta$$

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 2TP / (2TP + FP + FN)$$

Example 1 — Disease Detection

100 people are tested. 50 actually have the disease and 50 do not. The model correctly detects 40 diseased people, misses 10 diseased people, correctly identifies 45 healthy people, and incorrectly labels 5 healthy people as diseased.

	Predicted Disease	Predicted No Disease
Actual Disease	40 (TP)	10 (FN)
Actual No Disease	5 (FP)	45 (TN)

Step 1 — Identify TP, TN, FP and FN

From the matrix: **TP = 40, TN = 45, FP = 5, FN = 10.**

Check actual positives: $TP + FN = 40 + 10 = 50$. Check actual negatives: $TN + FP = 45 + 5 = 50$. Total = 100.

1. Accuracy

Formula: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Substitute: Accuracy = $(40 + 45) / 100$

Calculate: Accuracy = $85 / 100 = 0.8500$

Answer: Accuracy = 85.00%

Meaning: The percentage of all predictions that are correct.

2. Error Rate

Formula: Error Rate = $(FP + FN) / \text{Total} = 1 - \text{Accuracy}$

Substitute: Error Rate = $(5 + 10) / 100$

Calculate: Error Rate = $15 / 100 = 0.1500$

Answer: Error Rate = 15.00%

Meaning: The percentage of all predictions that are incorrect.

3. Precision (PPV)

Formula: Precision = $TP / (TP + FP)$

Substitute: Precision = $40 / (40 + 5)$

Calculate: Precision = $40 / 45 = 0.8889$

Answer: Precision = 88.89%

Meaning: Among cases predicted positive, how many are actually positive.

4. Recall / Sensitivity / TPR

Formula: Recall = $TP / (TP + FN)$

Substitute: Recall = $40 / (40 + 10)$

Calculate: Recall = $40 / 50 = 0.8000$

Answer: Recall = 80.00%

Meaning: Among cases that are actually positive, how many the model correctly finds.

5. Specificity / TNR

Formula: Specificity = $TN / (TN + FP)$

Substitute: Specificity = $45 / (45 + 5)$

Calculate: Specificity = $45 / 50 = 0.9000$

Answer: Specificity = 90.00%

Meaning: Among cases that are actually negative, how many the model correctly identifies.

6. False Positive Rate (FPR), α

Formula: $FPR = FP / (FP + TN) = 1 - \text{Specificity}$

Substitute: FPR = $5 / (5 + 45)$

Calculate: FPR = $5 / 50 = 0.1000$

Answer: FPR = 10.00%

Meaning: Among actual negatives, how many are incorrectly predicted positive.

7. False Negative Rate (FNR), β

Formula: $FNR = FN / (FN + TP) = 1 - \text{Sensitivity}$

Substitute: FNR = $10 / (10 + 40)$

Calculate: FNR = $10 / 50 = 0.2000$

Answer: FNR = 20.00%

Meaning: Among actual positives, how many are incorrectly predicted negative.

8. Power

Formula: Power = Sensitivity = TPR = $1 - \beta$

Substitute: Power = $1 - 0.2000$

Calculate: Power = 0.8000

Answer: Power = 80.00%

Meaning: The probability of correctly detecting an actual positive case.

9. F1 Score

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitute: F1 = $2 \times (0.8889 \times 0.8000) / (0.8889 + 0.8000)$

Calculate: F1 = 0.8421

Answer: F1 Score = 84.21%

Meaning: A single score that balances precision and recall.

Example 2 — Spam Email

200 emails are classified. 80 are actually spam and 120 are actually not spam. The model correctly identifies 70 spam emails, misses 10 spam emails, incorrectly marks 20 normal emails as spam, and correctly identifies 100 normal emails.

	Predicted Spam	Predicted Not Spam
Actual Spam	70 (TP)	10 (FN)
Actual Not Spam	20 (FP)	100 (TN)

Step 1 — Identify TP, TN, FP and FN

From the matrix: **TP = 70, TN = 100, FP = 20, FN = 10.**

Check actual positives: $TP + FN = 70 + 10 = 80$. Check actual negatives: $TN + FP = 100 + 20 = 120$. Total = 200.

1. Accuracy

Formula: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Substitute: Accuracy = $(70 + 100) / 200$

Calculate: Accuracy = $170 / 200 = 0.8500$

Answer: Accuracy = 85.00%

Meaning: The percentage of all predictions that are correct.

2. Error Rate

Formula: Error Rate = $(FP + FN) / \text{Total} = 1 - \text{Accuracy}$

Substitute: Error Rate = $(20 + 10) / 200$

Calculate: Error Rate = $30 / 200 = 0.1500$

Answer: Error Rate = 15.00%

Meaning: The percentage of all predictions that are incorrect.

3. Precision (PPV)

Formula: Precision = $TP / (TP + FP)$

Substitute: Precision = $70 / (70 + 20)$

Calculate: Precision = $70 / 90 = 0.7778$

Answer: Precision = 77.78%

Meaning: Among cases predicted positive, how many are actually positive.

4. Recall / Sensitivity / TPR

Formula: Recall = $TP / (TP + FN)$

Substitute: Recall = $70 / (70 + 10)$

Calculate: Recall = $70 / 80 = 0.8750$

Answer: Recall = 87.50%

Meaning: Among cases that are actually positive, how many the model correctly finds.

5. Specificity / TNR

Formula: Specificity = $TN / (TN + FP)$

Substitute: Specificity = $100 / (100 + 20)$

Calculate: Specificity = $100 / 120 = 0.8333$

Answer: Specificity = 83.33%

Meaning: Among cases that are actually negative, how many the model correctly identifies.

6. False Positive Rate (FPR), α

Formula: $FPR = FP / (FP + TN) = 1 - \text{Specificity}$

Substitute: FPR = $20 / (20 + 100)$

Calculate: FPR = $20 / 120 = 0.1667$

Answer: FPR = 16.67%

Meaning: Among actual negatives, how many are incorrectly predicted positive.

7. False Negative Rate (FNR), β

Formula: $FNR = FN / (FN + TP) = 1 - \text{Sensitivity}$

Substitute: FNR = $10 / (10 + 70)$

Calculate: FNR = $10 / 80 = 0.1250$

Answer: FNR = 12.50%

Meaning: Among actual positives, how many are incorrectly predicted negative.

8. Power

Formula: Power = Sensitivity = TPR = $1 - \beta$

Substitute: Power = $1 - 0.1250$

Calculate: Power = 0.8750

Answer: Power = 87.50%

Meaning: The probability of correctly detecting an actual positive case.

9. F1 Score

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitute: F1 = $2 \times (0.7778 \times 0.8750) / (0.7778 + 0.8750)$

Calculate: F1 = 0.8235

Answer: F1 Score = 82.35%

Meaning: A single score that balances precision and recall.

Example 3 — Loan Default

500 customers are considered. Positive = Will Default. 100 customers actually defaulted and 400 did not. The model correctly identifies 80 defaulters, misses 20 defaulters, incorrectly predicts default for 50 non-defaulters, and correctly predicts no default for 350.

	Predicted Default	Predicted No Default
Actual Default	80 (TP)	20 (FN)
Actual No Default	50 (FP)	350 (TN)

Step 1 — Identify TP, TN, FP and FN

From the matrix: **TP = 80, TN = 350, FP = 50, FN = 20.**

Check actual positives: $TP + FN = 80 + 20 = 100$. Check actual negatives: $TN + FP = 350 + 50 = 400$. Total = 500.

1. Accuracy

Formula: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Substitute: Accuracy = $(80 + 350) / 500$

Calculate: Accuracy = $430 / 500 = 0.8600$

Answer: Accuracy = 86.00%

Meaning: The percentage of all predictions that are correct.

2. Error Rate

Formula: Error Rate = $(FP + FN) / Total = 1 - Accuracy$

Substitute: Error Rate = $(50 + 20) / 500$

Calculate: Error Rate = $70 / 500 = 0.1400$

Answer: Error Rate = 14.00%

Meaning: The percentage of all predictions that are incorrect.

3. Precision (PPV)

Formula: Precision = $TP / (TP + FP)$

Substitute: Precision = $80 / (80 + 50)$

Calculate: Precision = $80 / 130 = 0.6154$

Answer: Precision = 61.54%

Meaning: Among cases predicted positive, how many are actually positive.

4. Recall / Sensitivity / TPR

Formula: Recall = $TP / (TP + FN)$

Substitute: Recall = $80 / (80 + 20)$

Calculate: Recall = $80 / 100 = 0.8000$

Answer: Recall = 80.00%

Meaning: Among cases that are actually positive, how many the model correctly finds.

5. Specificity / TNR

Formula: Specificity = $TN / (TN + FP)$

Substitute: Specificity = $350 / (350 + 50)$

Calculate: Specificity = $350 / 400 = 0.8750$

Answer: Specificity = 87.50%

Meaning: Among cases that are actually negative, how many the model correctly identifies.

6. False Positive Rate (FPR), α

Formula: $FPR = FP / (FP + TN) = 1 - \text{Specificity}$

Substitute: FPR = $50 / (50 + 350)$

Calculate: FPR = $50 / 400 = 0.1250$

Answer: FPR = 12.50%

Meaning: Among actual negatives, how many are incorrectly predicted positive.

7. False Negative Rate (FNR), β

Formula: $FNR = FN / (FN + TP) = 1 - \text{Sensitivity}$

Substitute: FNR = $20 / (20 + 80)$

Calculate: FNR = $20 / 100 = 0.2000$

Answer: FNR = 20.00%

Meaning: Among actual positives, how many are incorrectly predicted negative.

8. Power

Formula: Power = Sensitivity = TPR = $1 - \beta$

Substitute: Power = $1 - 0.2000$

Calculate: Power = 0.8000

Answer: Power = 80.00%

Meaning: The probability of correctly detecting an actual positive case.

9. F1 Score

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitute: F1 = $2 \times (0.6154 \times 0.8000) / (0.6154 + 0.8000)$

Calculate: F1 = 0.6957

Answer: F1 Score = 69.57%

Meaning: A single score that balances precision and recall.

Example 4 — Fraud Detection

1,000 transactions are classified. 100 are actually fraudulent and 900 are genuine. The model catches 90 frauds, misses 10 frauds, falsely flags 45 genuine transactions, and correctly clears 855 genuine transactions.

	Predicted Fraud	Predicted Genuine
Actual Fraud	90 (TP)	10 (FN)
Actual Genuine	45 (FP)	855 (TN)

Step 1 — Identify TP, TN, FP and FN

From the matrix: **TP = 90, TN = 855, FP = 45, FN = 10.**

Check actual positives: $TP + FN = 90 + 10 = 100$. Check actual negatives: $TN + FP = 855 + 45 = 900$. Total = 1000.

1. Accuracy

Formula: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Substitute: Accuracy = $(90 + 855) / 1000$

Calculate: Accuracy = $945 / 1000 = 0.9450$

Answer: Accuracy = 94.50%

Meaning: The percentage of all predictions that are correct.

2. Error Rate

Formula: Error Rate = $(FP + FN) / \text{Total} = 1 - \text{Accuracy}$

Substitute: Error Rate = $(45 + 10) / 1000$

Calculate: Error Rate = $55 / 1000 = 0.0550$

Answer: Error Rate = 5.50%

Meaning: The percentage of all predictions that are incorrect.

3. Precision (PPV)

Formula: Precision = $TP / (TP + FP)$

Substitute: Precision = $90 / (90 + 45)$

Calculate: Precision = $90 / 135 = 0.6667$

Answer: Precision = 66.67%

Meaning: Among cases predicted positive, how many are actually positive.

4. Recall / Sensitivity / TPR

Formula: Recall = $TP / (TP + FN)$

Substitute: Recall = $90 / (90 + 10)$

Calculate: Recall = $90 / 100 = 0.9000$

Answer: Recall = 90.00%

Meaning: Among cases that are actually positive, how many the model correctly finds.

5. Specificity / TNR

Formula: Specificity = $TN / (TN + FP)$

Substitute: Specificity = $855 / (855 + 45)$

Calculate: Specificity = $855 / 900 = 0.9500$

Answer: Specificity = 95.00%

Meaning: Among cases that are actually negative, how many the model correctly identifies.

6. False Positive Rate (FPR), α

Formula: $FPR = FP / (FP + TN) = 1 - \text{Specificity}$

Substitute: $FPR = 45 / (45 + 855)$

Calculate: $FPR = 45 / 900 = 0.0500$

Answer: FPR = 5.00%

Meaning: Among actual negatives, how many are incorrectly predicted positive.

7. False Negative Rate (FNR), β

Formula: $FNR = FN / (FN + TP) = 1 - \text{Sensitivity}$

Substitute: $FNR = 10 / (10 + 90)$

Calculate: $FNR = 10 / 100 = 0.1000$

Answer: FNR = 10.00%

Meaning: Among actual positives, how many are incorrectly predicted negative.

8. Power

Formula: Power = Sensitivity = $TPR = 1 - \beta$

Substitute: Power = $1 - 0.1000$

Calculate: Power = 0.9000

Answer: Power = 90.00%

Meaning: The probability of correctly detecting an actual positive case.

9. F1 Score

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitute: $F1 = 2 \times (0.6667 \times 0.9000) / (0.6667 + 0.9000)$

Calculate: $F1 = 0.7660$

Answer: F1 Score = 76.60%

Meaning: A single score that balances precision and recall.

Example 5 — Student Pass Prediction

200 students are classified. 120 actually pass and 80 actually fail. The model predicts 105 pass correctly, predicts 15 as fail even though they pass, predicts 10 as pass even though they fail, and predicts 70 fail correctly.

	Predicted Pass	Predicted Fail
Actual Pass	105 (TP)	15 (FN)
Actual Fail	10 (FP)	70 (TN)

Step 1 — Identify TP, TN, FP and FN

From the matrix: **TP = 105, TN = 70, FP = 10, FN = 15.**

Check actual positives: $TP + FN = 105 + 15 = 120$. Check actual negatives: $TN + FP = 70 + 10 = 80$. Total = 200.

1. Accuracy

Formula: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Substitute: Accuracy = $(105 + 70) / 200$

Calculate: Accuracy = $175 / 200 = 0.8750$

Answer: Accuracy = 87.50%

Meaning: The percentage of all predictions that are correct.

2. Error Rate

Formula: Error Rate = $(FP + FN) / \text{Total} = 1 - \text{Accuracy}$

Substitute: Error Rate = $(10 + 15) / 200$

Calculate: Error Rate = $25 / 200 = 0.1250$

Answer: Error Rate = 12.50%

Meaning: The percentage of all predictions that are incorrect.

3. Precision (PPV)

Formula: Precision = $TP / (TP + FP)$

Substitute: Precision = $105 / (105 + 10)$

Calculate: Precision = $105 / 115 = 0.9130$

Answer: Precision = 91.30%

Meaning: Among cases predicted positive, how many are actually positive.

4. Recall / Sensitivity / TPR

Formula: Recall = $TP / (TP + FN)$

Substitute: Recall = $105 / (105 + 15)$

Calculate: Recall = $105 / 120 = 0.8750$

Answer: Recall = 87.50%

Meaning: Among cases that are actually positive, how many the model correctly finds.

5. Specificity / TNR

Formula: Specificity = $TN / (TN + FP)$

Substitute: Specificity = $70 / (70 + 10)$

Calculate: Specificity = $70 / 80 = 0.8750$

Answer: Specificity = 87.50%

Meaning: Among cases that are actually negative, how many the model correctly identifies.

6. False Positive Rate (FPR), α

Formula: $FPR = FP / (FP + TN) = 1 - \text{Specificity}$

Substitute: FPR = $10 / (10 + 70)$

Calculate: FPR = $10 / 80 = 0.1250$

Answer: FPR = 12.50%

Meaning: Among actual negatives, how many are incorrectly predicted positive.

7. False Negative Rate (FNR), β

Formula: $FNR = FN / (FN + TP) = 1 - \text{Sensitivity}$

Substitute: FNR = $15 / (15 + 105)$

Calculate: FNR = $15 / 120 = 0.1250$

Answer: FNR = 12.50%

Meaning: Among actual positives, how many are incorrectly predicted negative.

8. Power

Formula: Power = Sensitivity = TPR = $1 - \beta$

Substitute: Power = $1 - 0.1250$

Calculate: Power = 0.8750

Answer: Power = 87.50%

Meaning: The probability of correctly detecting an actual positive case.

9. F1 Score

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitute: F1 = $2 \times (0.9130 \times 0.8750) / (0.9130 + 0.8750)$

Calculate: F1 = 0.8936

Answer: F1 Score = 89.36%

Meaning: A single score that balances precision and recall.

Final Quick Revision

Metric	Formula	What it tells you
Accuracy	$(TP + TN) / \text{Total}$	Overall correctness
Error Rate	$(FP + FN) / \text{Total}$	Overall mistakes
Precision	$TP / (TP + FP)$	How reliable positive predictions are
Recall / Sensitivity	$TP / (TP + FN)$	How many actual positives are found
Specificity	$TN / (TN + FP)$	How many actual negatives are found
FPR (α)	$FP / (FP + TN)$	False alarms among actual negatives
FNR (β)	$FN / (FN + TP)$	Missed positives among actual positives
Power	$1 - \beta$	Correctly detected positives
F1 Score	$2PR / (P + R)$	Balance of precision and recall

Exam shortcut: If you are given a confusion matrix, first write down TP, TN, FP, FN. Then select the required formula and substitute the numbers. Do not swap FP with FN.